NILAI UNIVERSITY

BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION
DECEMBER 2025

Course: EC3123 (Operating Systems Principles)

Lecturer: Mohd Shahdi Bin Ahmad

Time: 2.00PM – 5.00PM
(3 hours)

Date: 12 February 2026

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **2** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1	a.	Differentiate between a program and a process by explaining **THREE (3)** key differences.	**(6 MARKS)**
	b.	Explain the **FIVE (5)** process states that occur during program execution.	**(10 MARKS)**
	c.	Construct a process creation flow diagram, clearly labelling each stage and showing the sequence and relationship of system calls involved.	**(7 MARKS)**
	d.	Define the term multithreading.	**(2 MARKS)**
2	a.	Explain the fundamental concepts of static memory allocation and dynamic memory allocation.	**(4 MARKS)**
	b.	Using a suitable diagram, demonstrate how the swapping mechanism operates in memory management.	**(8 MARKS)**
	c.	Using diagrams, demonstrate single-partition and multiple-partition memory allocation techniques in contiguous memory management.	**(8 MARKS)**
	d.	List **FIVE (5)** segments that make up a logical address space.	**(5 MARKS)**
3	a.	State the main objectives of I/O software.	**(5 MARKS)**
	b.	List the layers that form the I/O software architecture.	**(5 MARKS)**
	c.	Draw and label a single-bus architecture used in memory-mapped I/O.	**(6 MARKS)**
	d.	Draw and label a dual-bus memory architecture for memory-mapped I/O.	**(7 MARKS)**
	e.	Provide **TWO (2)** examples of human-readable I/O devices.	**(2 MARKS)**

4 a. Write appropriate Linux commands to perform the following tasks:

i. Create a directory named "Computing". **(2 MARKS)**

ii. Delete a non-empty directory named "Department". **(2 MARKS)**

b. Using the information given in Table 1, answer the following:

No.	Group Name	Group ID	Username	User ID
1	Designer	3000	Jeff	3001
2	Programmer	4000	Lim	4001
3	Administrator	5000	Hamse	5001
4	Administrator	5000	Remy	5002
5	Finance	6000	Mary	6001
6	Finance	6000	Nani	6002

Table 1

i. Write Linux commands to create all user groups listed. **(8 MARKS)**

ii. Write Linux commands to create all user accounts listed. **(12 MARKS)**

c. State the Linux command used to edit a file. **(1 MARK)**

-END OF QUESTION PAPER-